

The Internet protocols

Maria Kihl



LUND
UNIVERSITY

Book chapters

Forouzan: 1.3, 2.4, 2.5, 19, 20, 21.1, 23.1-3, 25.1-4

Kihl: 8.1-7

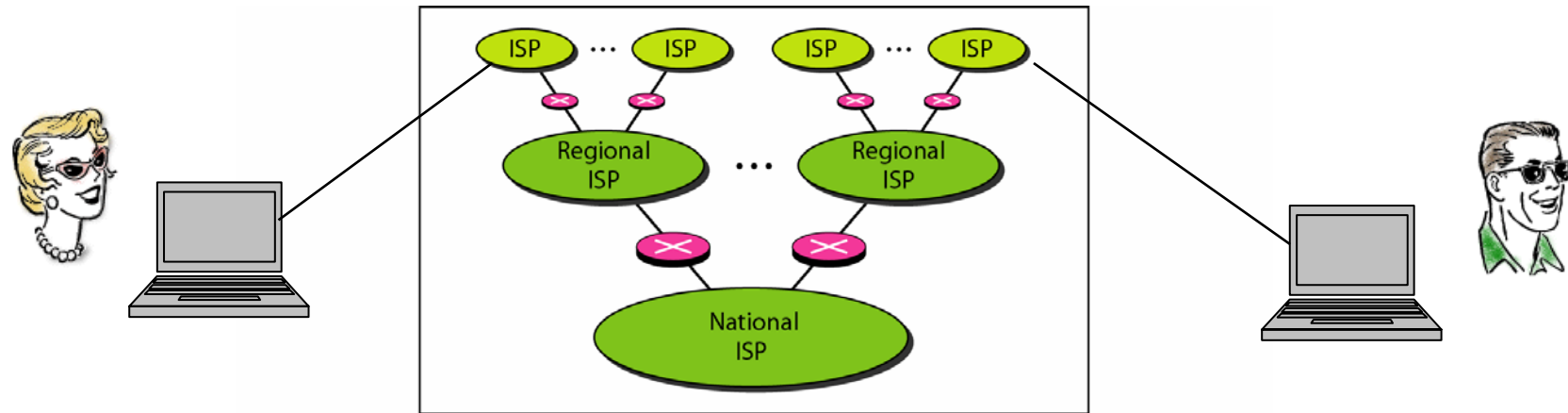
What is Internet?

- Internet consists of a number of networks that exchange data according to traffic agreements.
- Internet is partly controlled by the Internet Society (ISOC).
- All networks in Internet use the TCP/IP protocol suite.

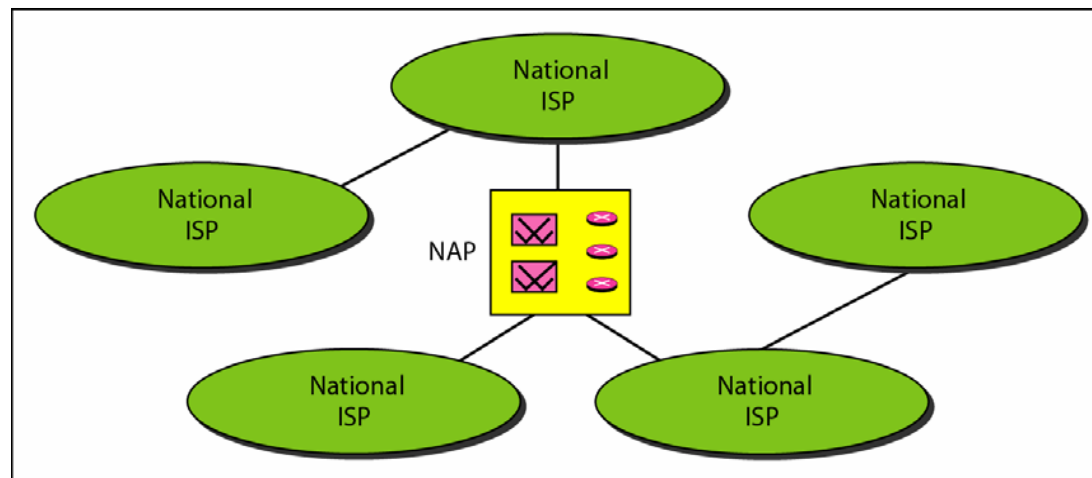
Network architecture

- Two types of networks in Internet: **Backbone networks** and **Access networks**.
- The end-user is connected to an access network.
- Today, the access networks and some of the backbone networks are owned by private so called **Internet Service Providers (ISPs)**.
- **Network Access Points, NAPs**, (or **Internet Exchanges, IX**) are switching stations that provide connection between backbone networks.

Organisation of Internet

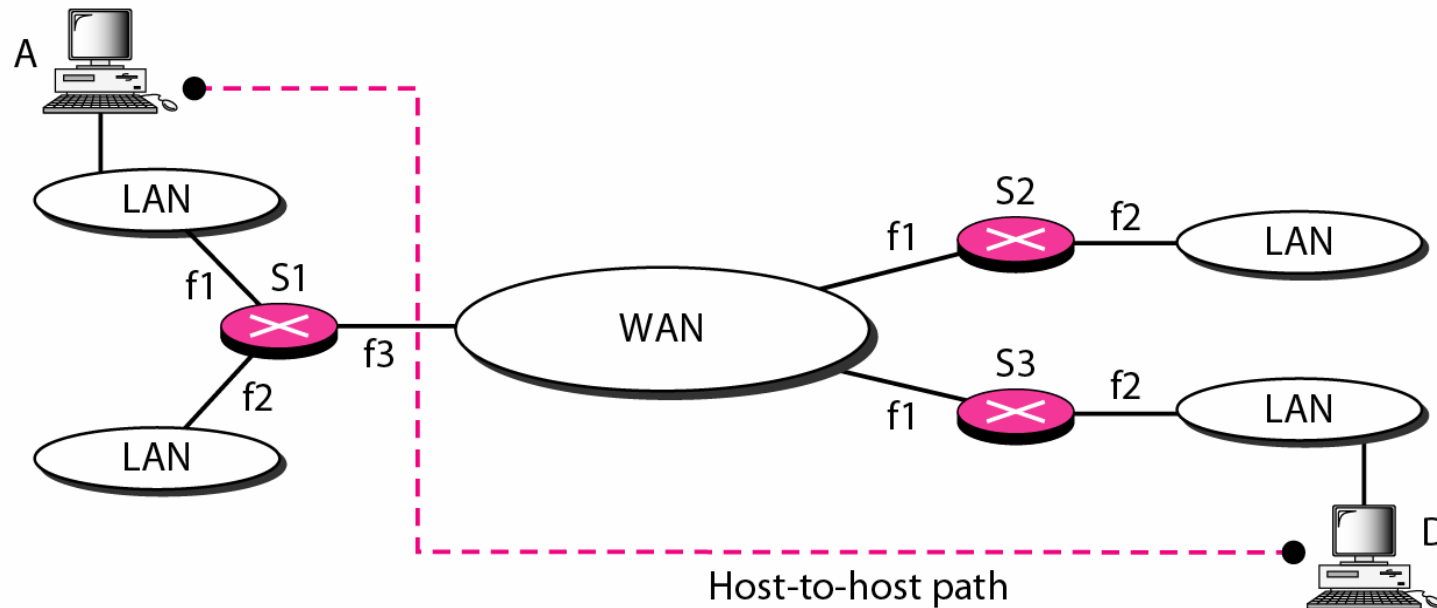


a. Structure of a national ISP



b. Interconnection of national ISPs

Internetwork



Internet?

All networks that are part of Internet have one thing in common:

They all use the same network protocol, **Internet Protocol (IP)**!

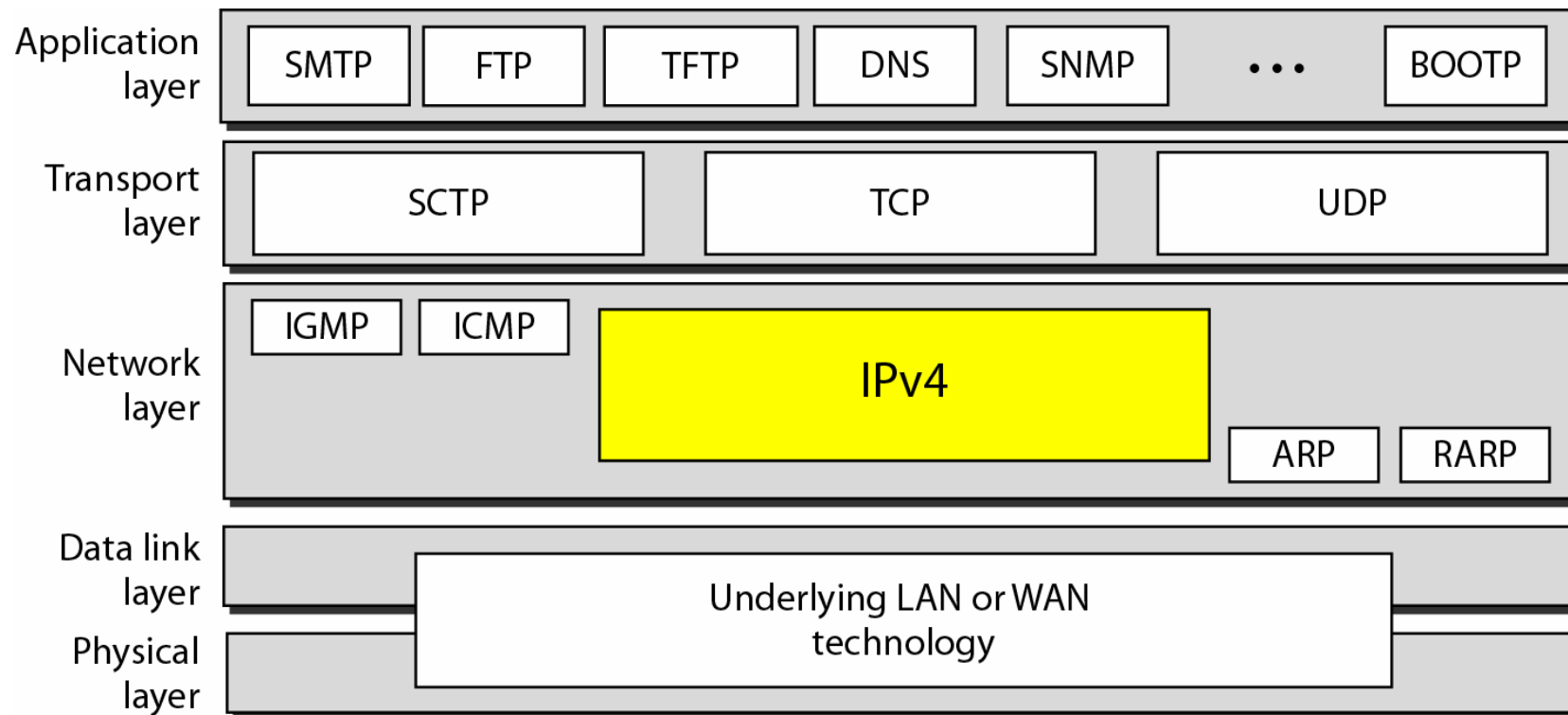
However, they do not need to use the same link protocols, which means that all kinds of networks can be part of Internet.

The protocol framework for Internet is called the **TCP/IP-model**.

Comparison with the OSI-model

OSI-model	TCP/IP-model
Application	Application
Presentation	
Session	
Transport	Transport
Network	Network
Link	Underlying network
Physical	

TCP/IP protocol suite



Why is this important?

- The data networks have been developed during more than 30 years.
- During this time, several networks have been standardized with different protocols on different layers.
- Since IP can be used on all types of networks, this has made Internet what it is today.
- All documentation about the Internet protocols is so called open access.

Internet protocol (IP)

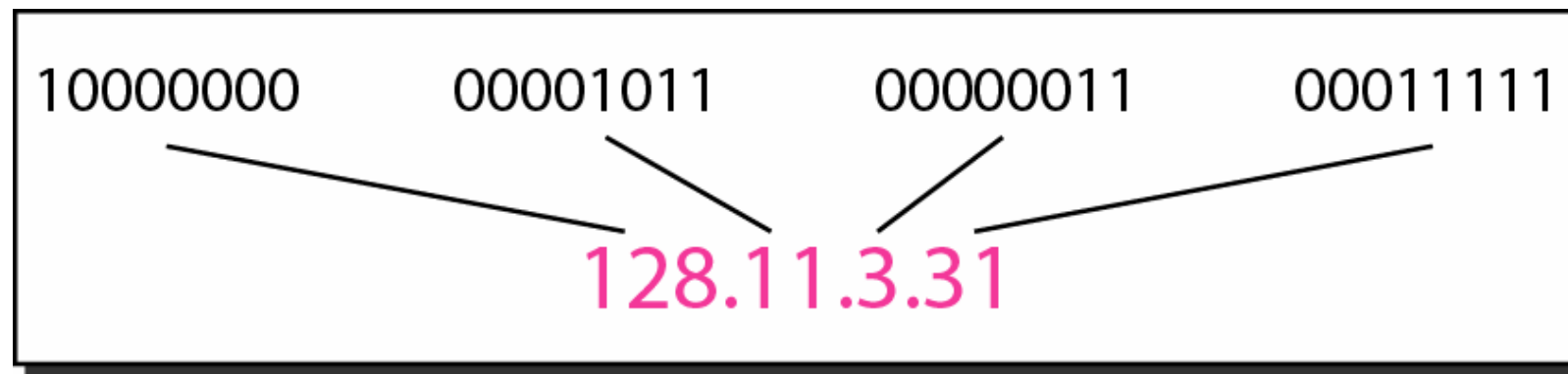
Two versions of IP:

- IPv4 is used today that will be the main focus in this course.
- IPv6 is an improved future network protocol, which we will only introduce in this course.

IP is a so called **Best effort** protocol, which means that it does not support any error control methods. Also, IP is based on **connectionless** transmission.

IP addresses (IPv4)

Every host and router that are connected to the Internet have a unique network address (IP address) that is 32 bits long.



The address is usually written in a dotted-decimal format.

IP addresses

The IPv4 address consists of two parts:

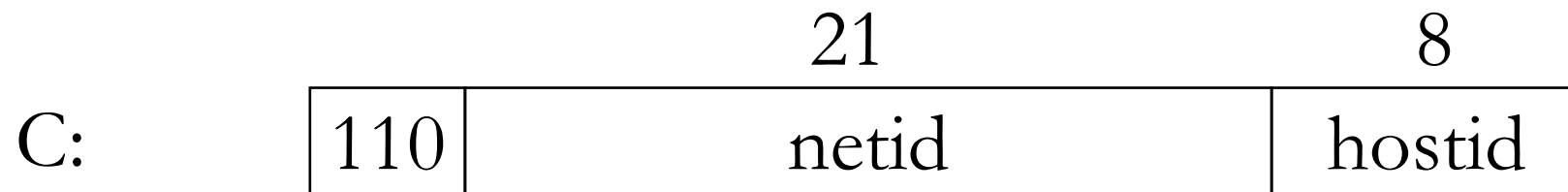
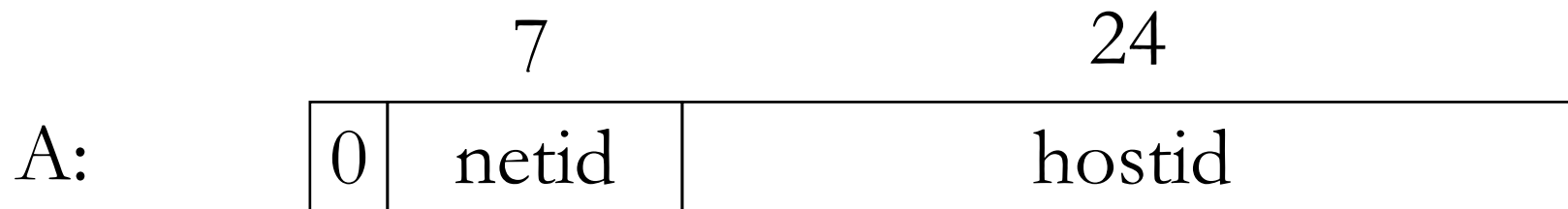
- **Netid** defines the network that the device is connected to.
- **Hostid** defines the device itself within this network.

There are two ways to define addresses:

- **Classful addressing**
- **Classless addressing**

Classful addressing

Five address classes are defined: A, B, C, (D, and E)



Netid and hostid in Classful addressing

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			

b. Dotted-decimal notation

 = netid

Classful addressing

The major problem with classful addressing is that an organization can only get a **block** of addresses.

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast

Therefore, classful addressing is now replaced with **classless addressing**.

Classless addressing

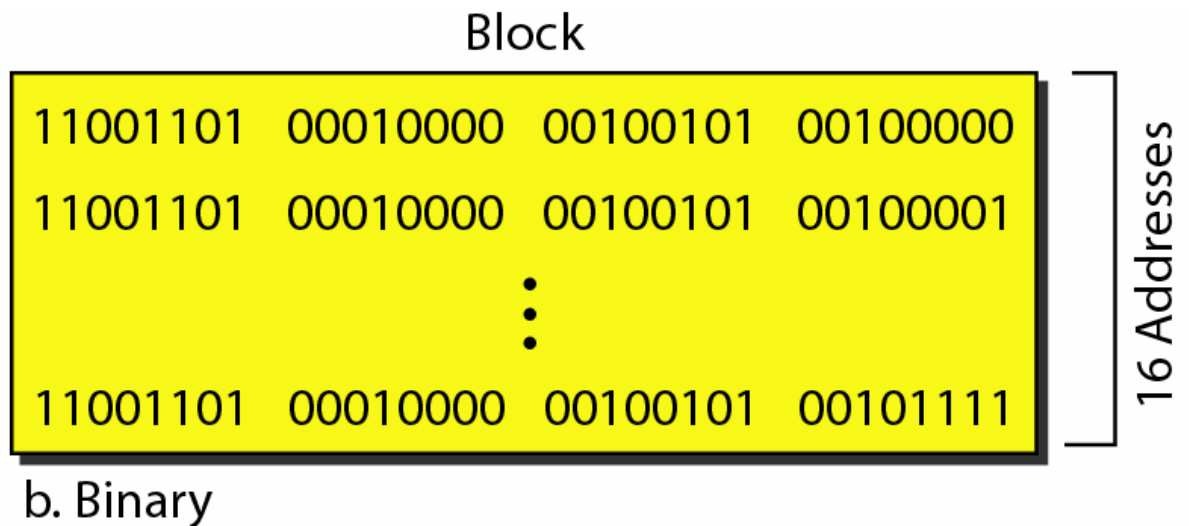
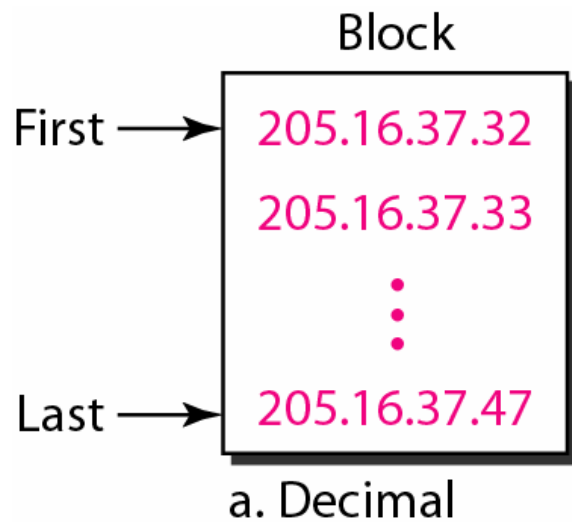
In classless addressing, an organization (small or large) is granted a block of addresses, with the following restrictions:

The addresses in a block must be contiguous.

The number of addresses in a block must be a power of 2 (1, 2, 4, 8, etc.).

The first address must be evenly divisible by the number of addresses.

Classless addressing, example



Mask

An address block can be defined by its **mask**.

A mask is a 32-bit number in which the n leftmost bits are 1s and the $32-n$ rightmost bits are 0s.

In IPv4 addressing, a block of addresses can be defined as:

$x.y.z.t / n$

in which $x.y.z.t$ defines one of the addresses and the $/n$ defines the mask.

The slash notation $/n$ is called **Classless Interdomain Routing (CIDR)**

Classless addressing, example

Address: 11011110 00010111 01000011 01000100

Mask: 11111111 11111111 11000000 00000000

Netid: 11011110 00010111 01000000 00000000

Hostid: 00000000 00000000 00000011 01000100

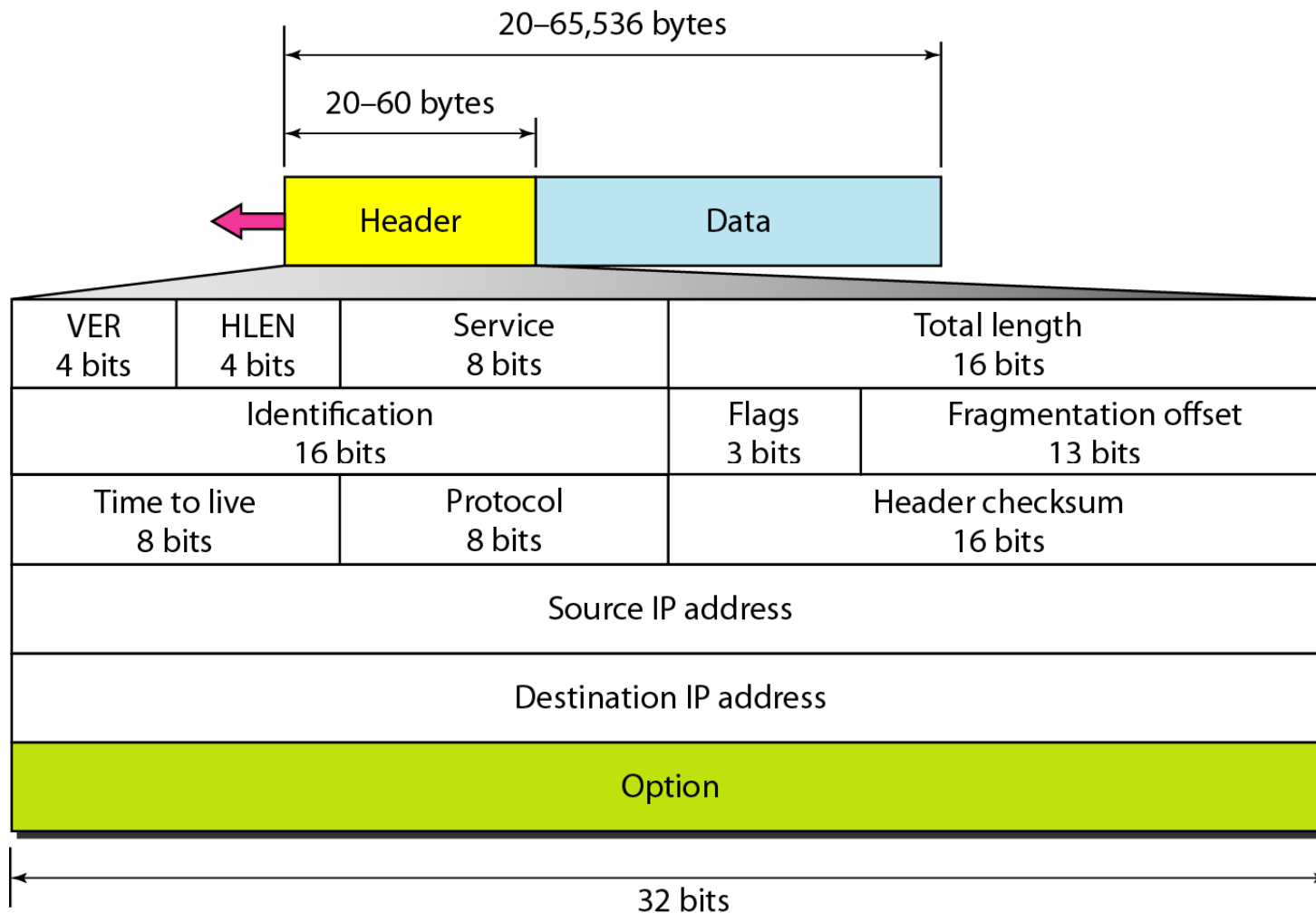
Decimal-dotted format: 232.23.67.68/18

Classless v. classful addressing

The classful addressing scheme can also be represented with masks:

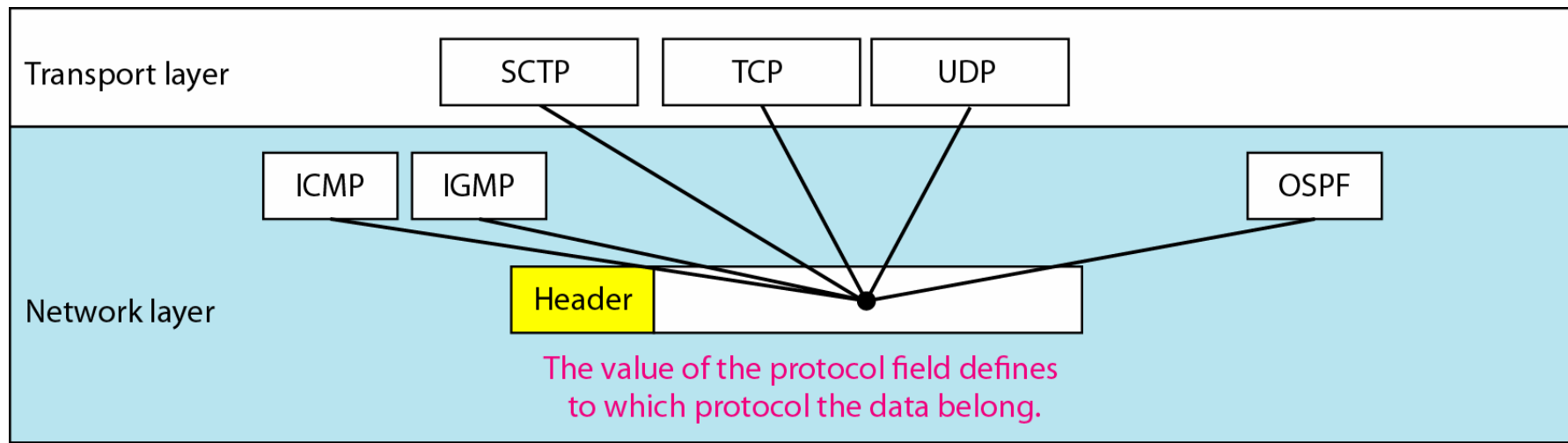
<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255 .0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255 .0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255 .0	/24

IPv4 datagram format



Encapsulated data

The protocol field contains information about what higher-level protocol that uses the services of IPv4.



Header checksum, example

The header contains a 16-bit checksum:

4	5	0	28	
1			0	0
4	17	0		
10.12.14.5				
12.6.7.9				

4, 5, and 0

28

1

0 and 0

4 and 17

0

10.12

14.5

12.6

7.9

Sum

Checksum

→

→

→

→

→

→

→

→

→

→

→

→

4	5	0	0
0	0	1	C
0	0	0	1
0	0	0	0
0	4	1	1
0	0	0	0
0	A	0	C
0	E	0	5
0	C	0	6
0	7	0	9
7	4	4	E
8	B	B	1

</

Reasons for IPv6

IPv4 has the following major problems:

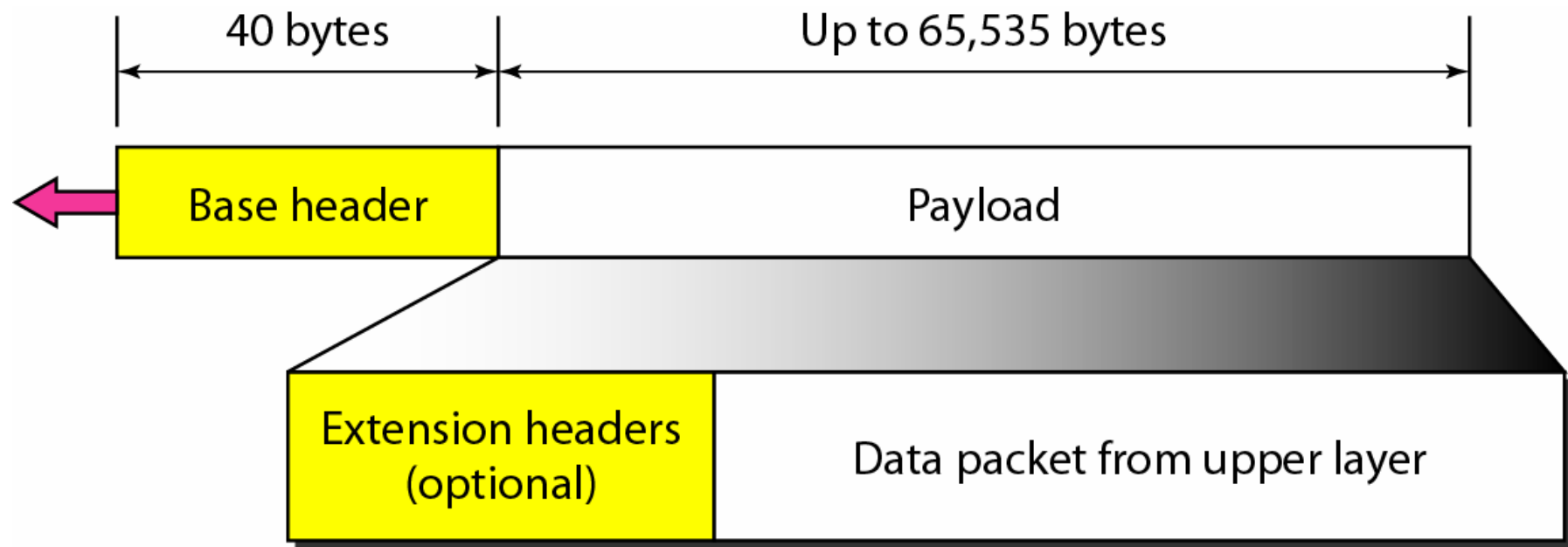
- The address space is too small.
- It was not designed for real-time applications.
- It has no support for encryption and authentication.

Therefore, **IPv6** was developed. However, the adoption of IPv6 has been slow.

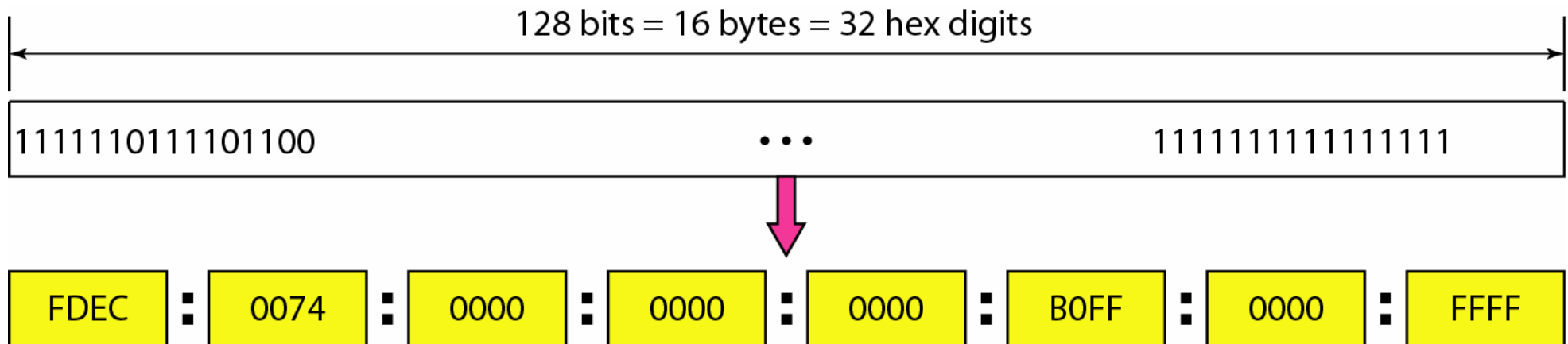
Some advantages with IPv6

- Larger address space: 128 bits long addresses.
- Better header format: The base header has a constant length (40 bytes). Options can be inserted, when needed.
- Support for more security: IPv6 has encryption and authentication options.
- Support for real-time applications: Special handling of datagram can be requested.

IPv6 packet format

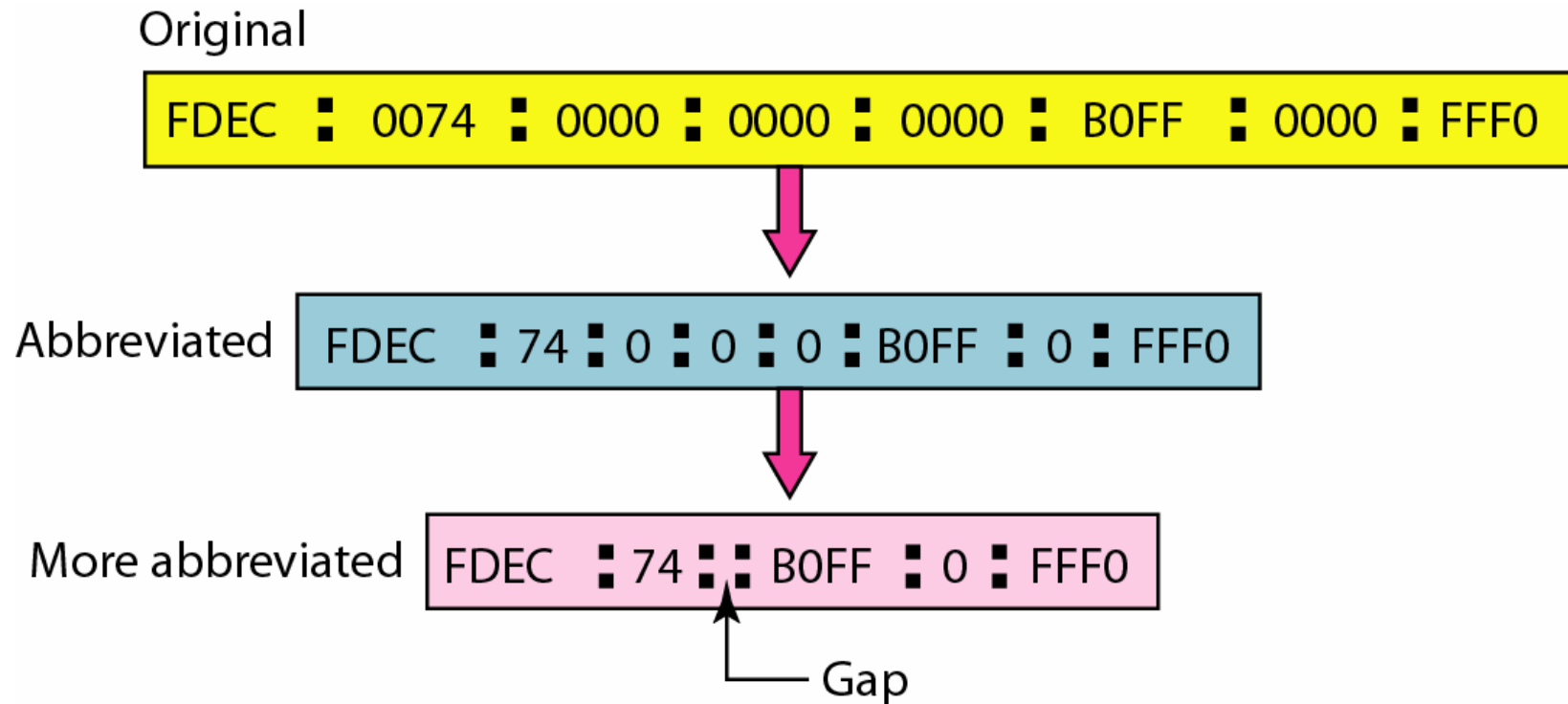


IPv6 addresses



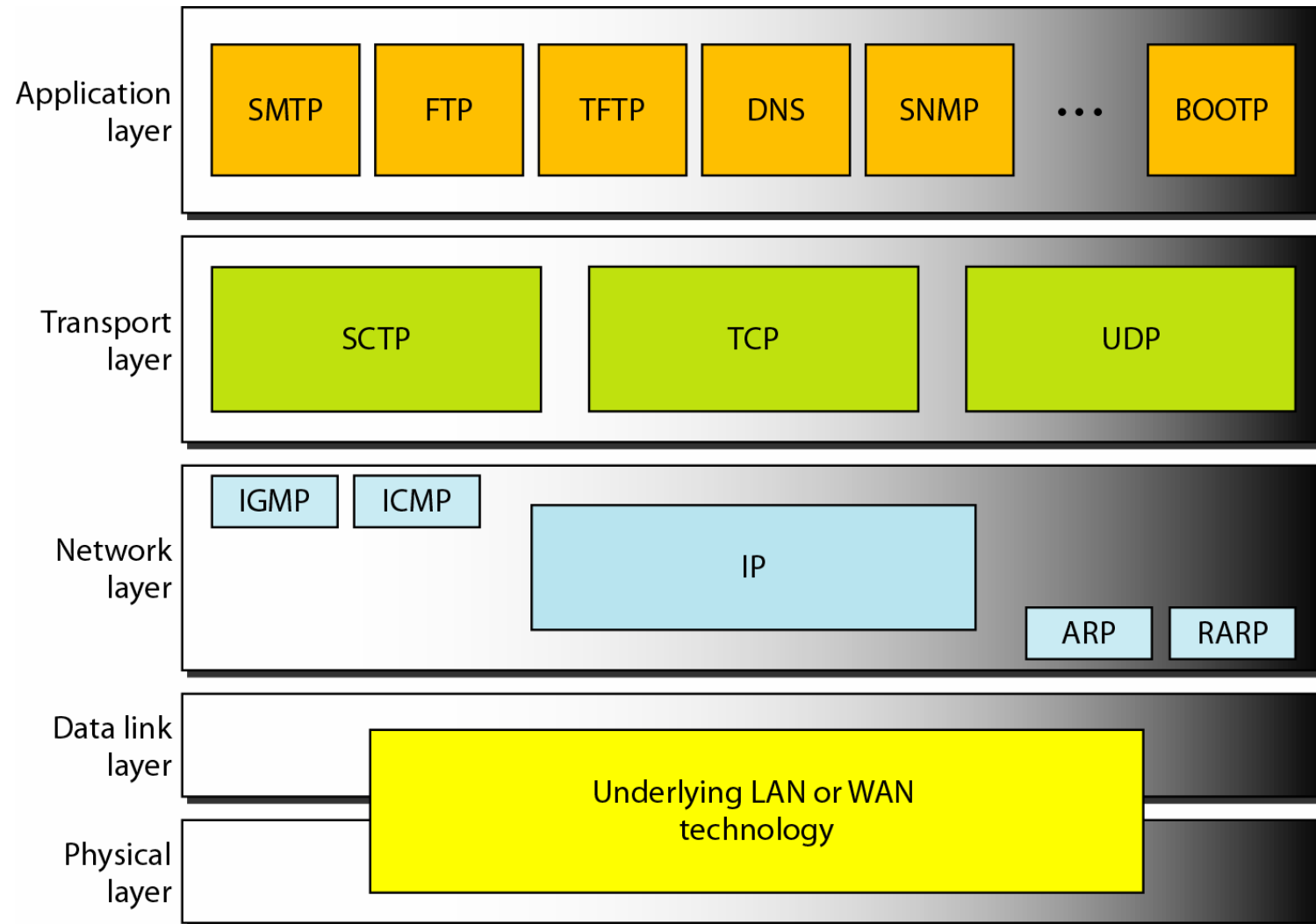
Hexadecimal colon notation

Abbreviated IPv6 addresses

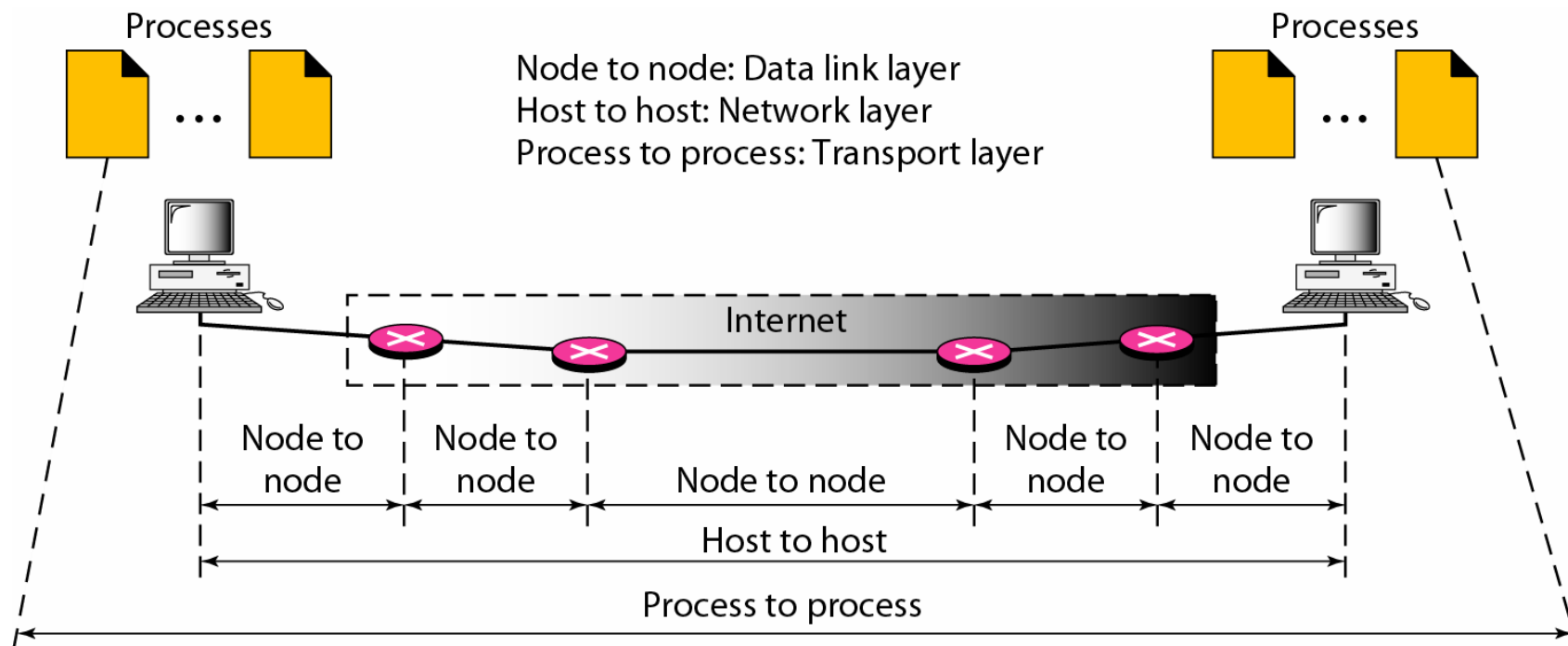


The removal of whole sections of zeros can only be done once per address.

Transport protocols in Internet



Process-to-process delivery



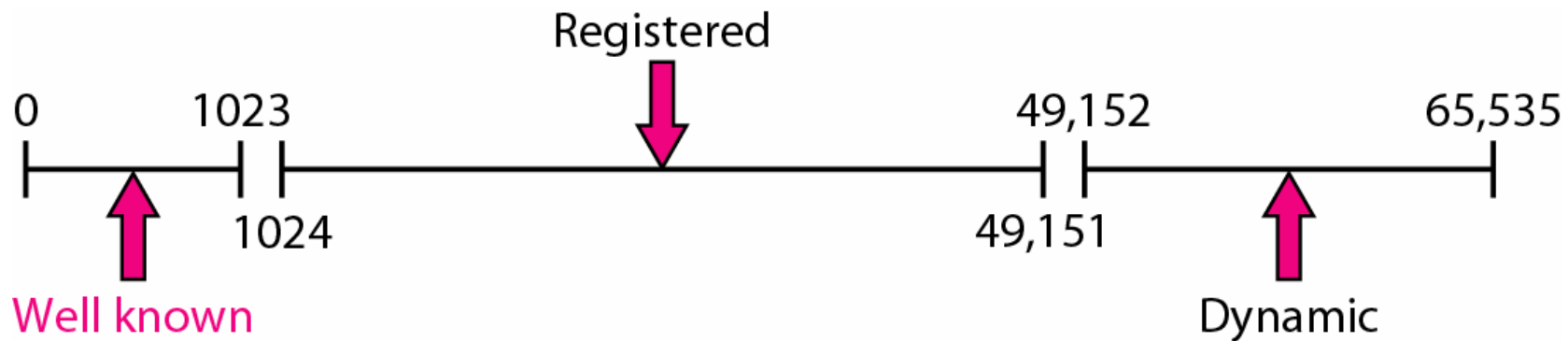
Port numbers

IANA has defined three types of port numbers:

- **Well-known ports** are assigned and controlled by IANA.
- **Registered ports** can be registered with IANA to prevent duplication.
- **Dynamic ports** are neither controlled nor registered, and can be used by any software. They are also called **ephemeral ports**.

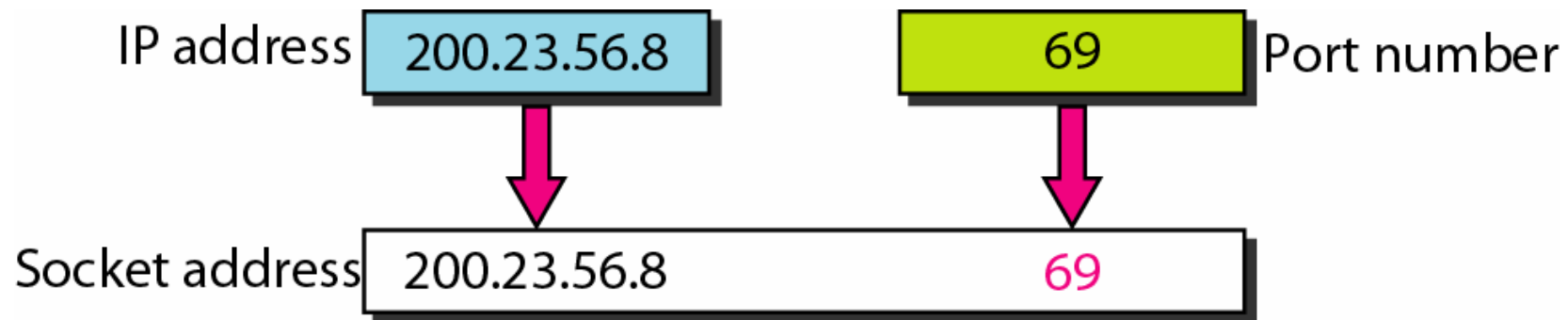
IANA ranges

The port addresses are within the following ranges:



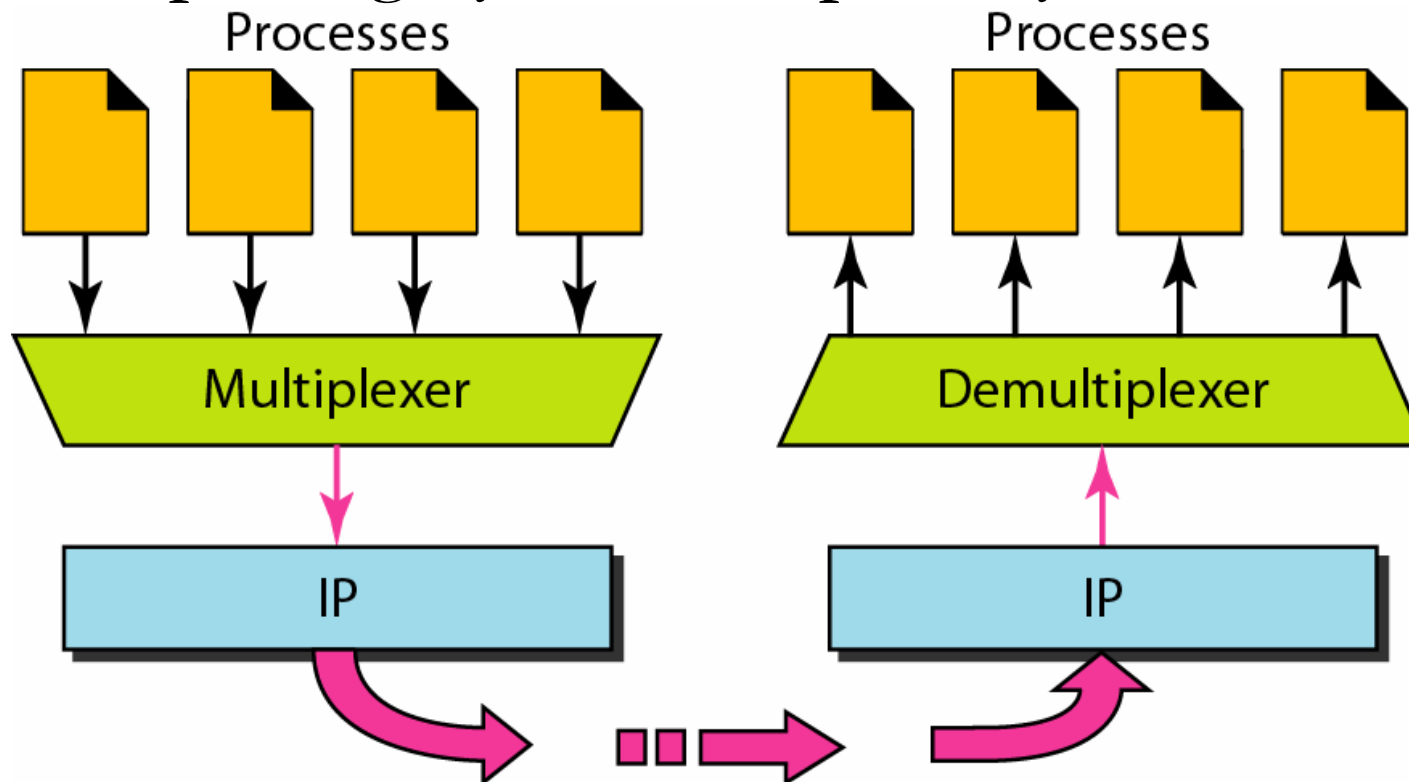
Socket address

The combination of an IP address and a port number is called a **socket address**.



Multiplexing and demultiplexing

The socket addresses allow multiplexing and demultiplexing by the transport layer:



User Datagram Protocol (UDP)

UDP is a **connectionless, unreliable** transport protocol. It does not add anything to services of IP except to provide process-to-process communication instead of host-to-host communication.

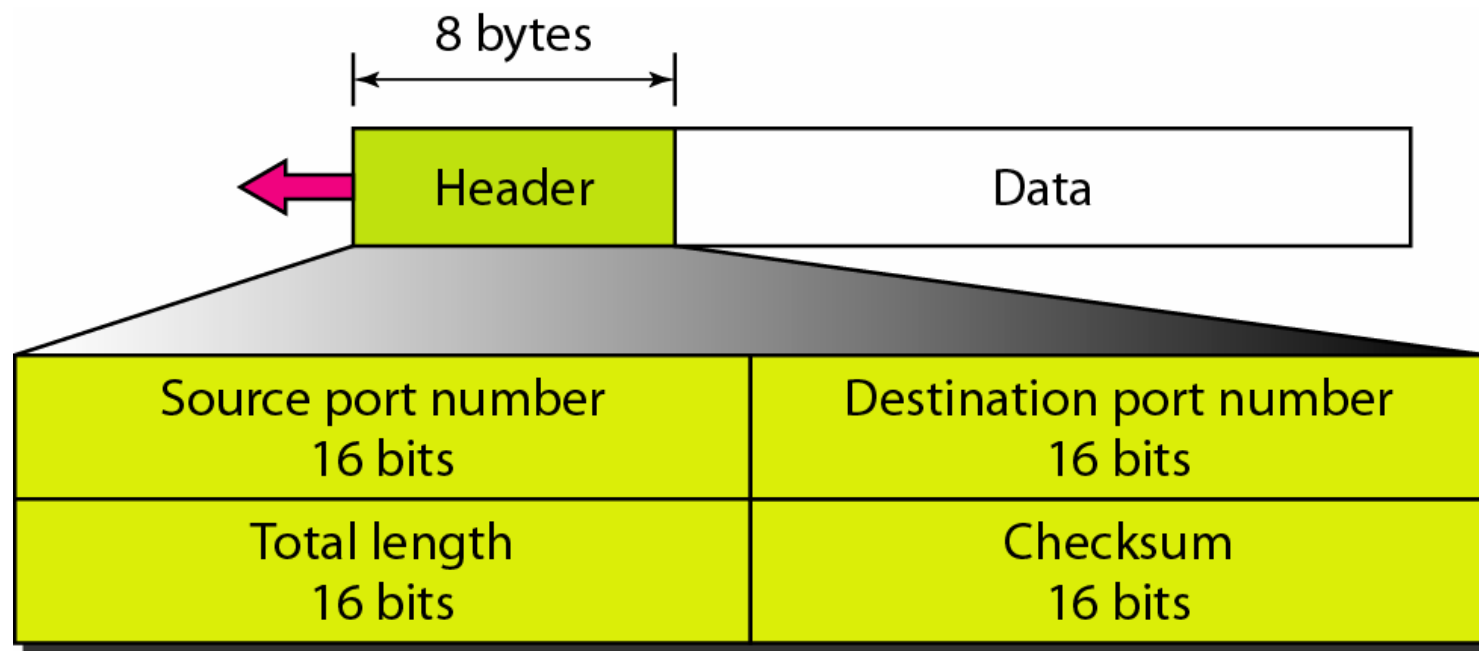
Advantage: It is very simple and adds a **minimum of overhead**.

Some applications that use UDP: Online games, IP-telephony.

Some well-known UDP ports

<i>Port</i>	<i>Protocol</i>	<i>Description</i>
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
53	Nameserver	Domain Name Service
67	BOOTPs	Server port to download bootstrap information
68	BOOTPc	Client port to download bootstrap information
69	TFTP	Trivial File Transfer Protocol
111	RPC	Remote Procedure Call
123	NTP	Network Time Protocol
161	SNMP	Simple Network Management Protocol
162	SNMP	Simple Network Management Protocol (trap)

UDP packet format



Transmission Control Protocol (TCP)

TCP is a **connection-oriented, reliable** transport protocol.

It creates a virtual connection between two processes.

Also, it provides flow and error control mechanisms.

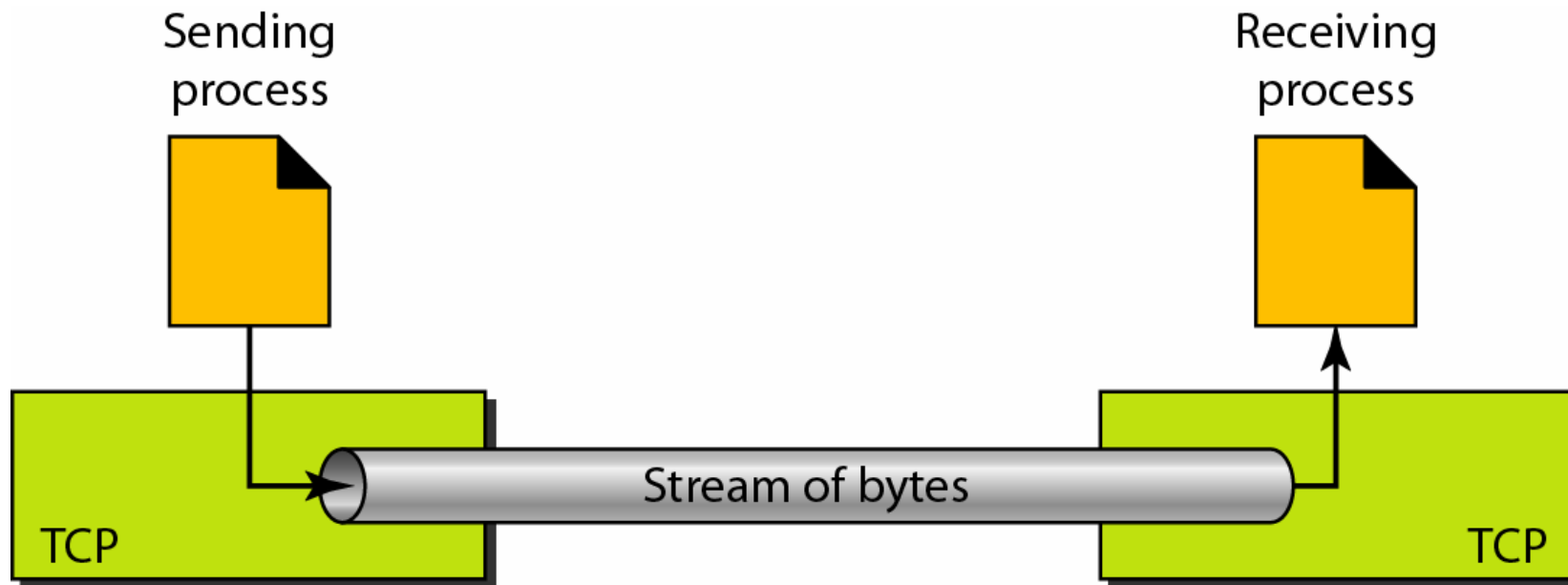
Some applications that use TCP: HTTP, File transfer, Email.

Some well-known TCP ports

<i>Port</i>	<i>Protocol</i>	<i>Description</i>
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FTP, Data	File Transfer Protocol (data connection)
21	FTP, Control	File Transfer Protocol (control connection)
23	TELNET	Terminal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol
111	RPC	Remote Procedure Call

Stream delivery

TCP allows the sending process to deliver data as a **stream of bytes** and allows the receiving process to obtain data as a stream of bytes.



TCP functions

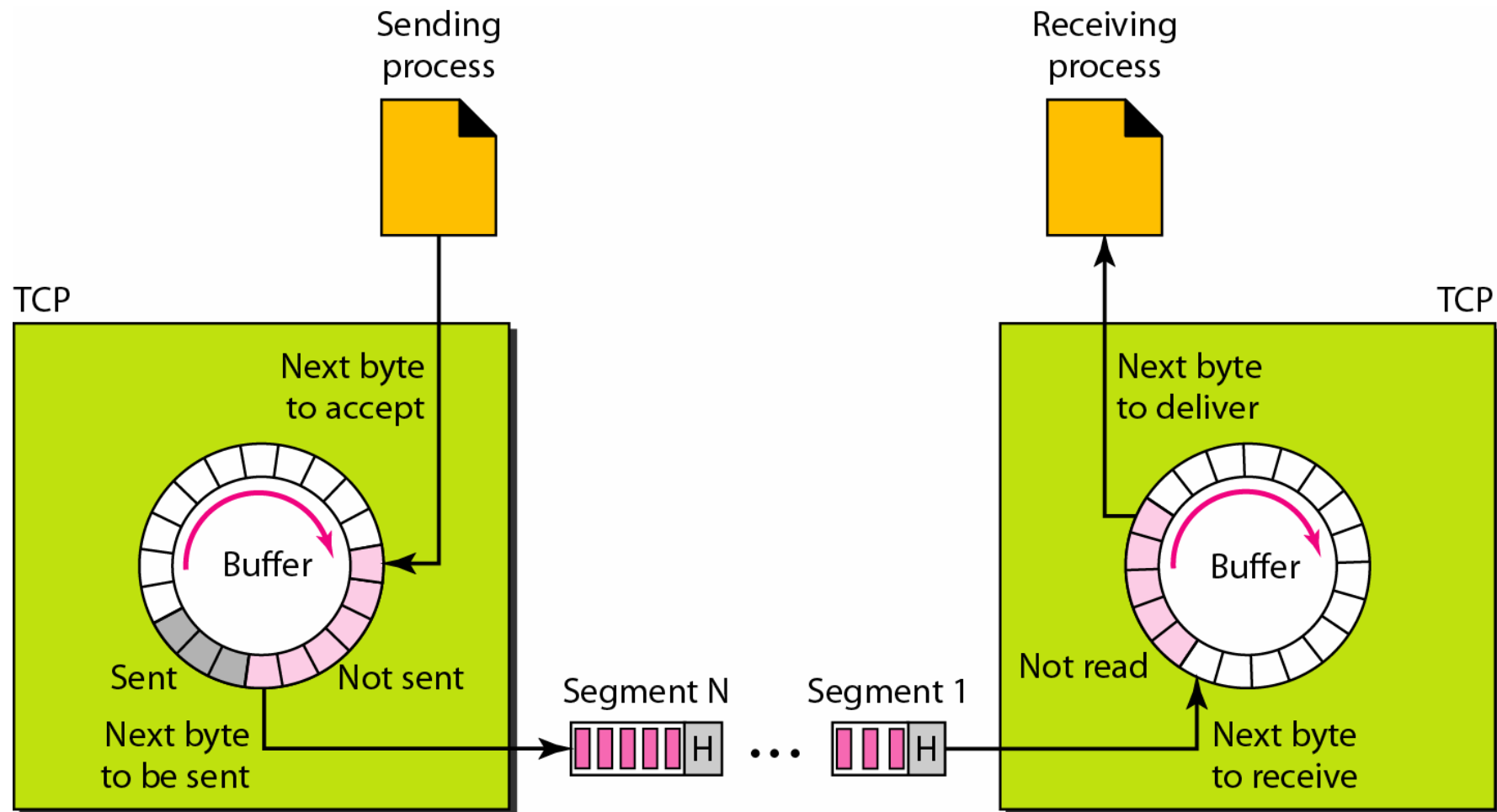
TCP groups a number of bytes together into a packet called a **segment**.

To allow for flow and error control, both the sender and receiver use **buffers**.

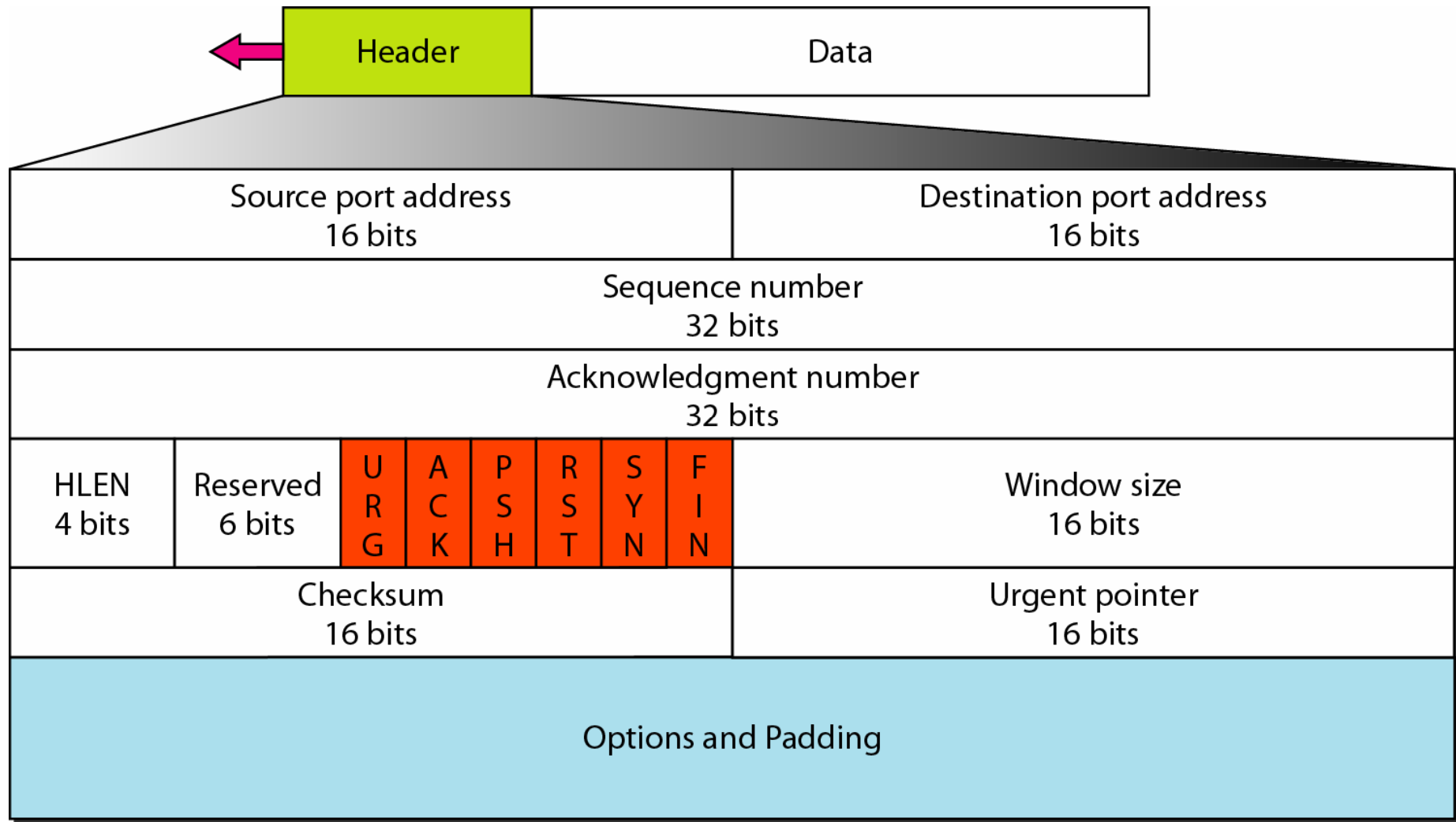
TCP implements a **Go-back-N protocol** in which the sequence number is the number of the first **byte** carried in the segment. The ACK contains the number of the next byte that the receiver expects to receive.

ACKs can be **piggybacked**.

TCP Segments, example

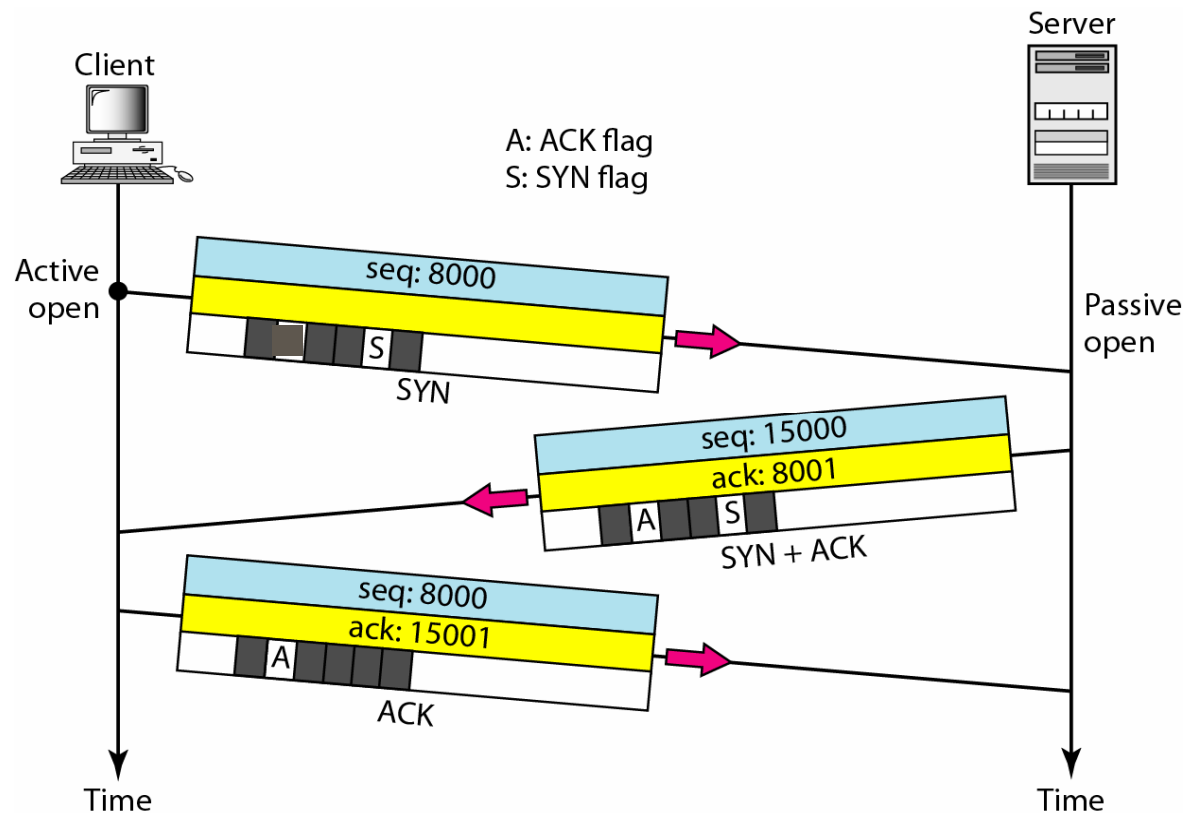


TCP segment format

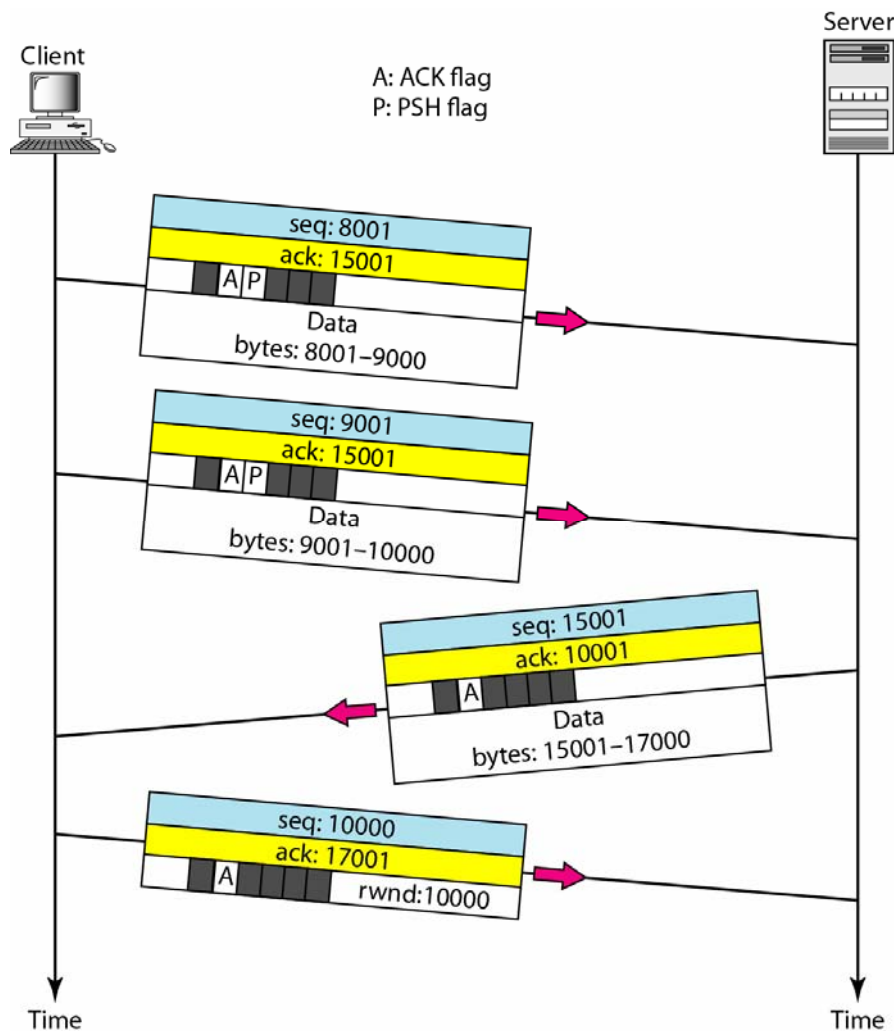


TCP connection establishment

The connection establishment in TCP is called **three-way handshake**.

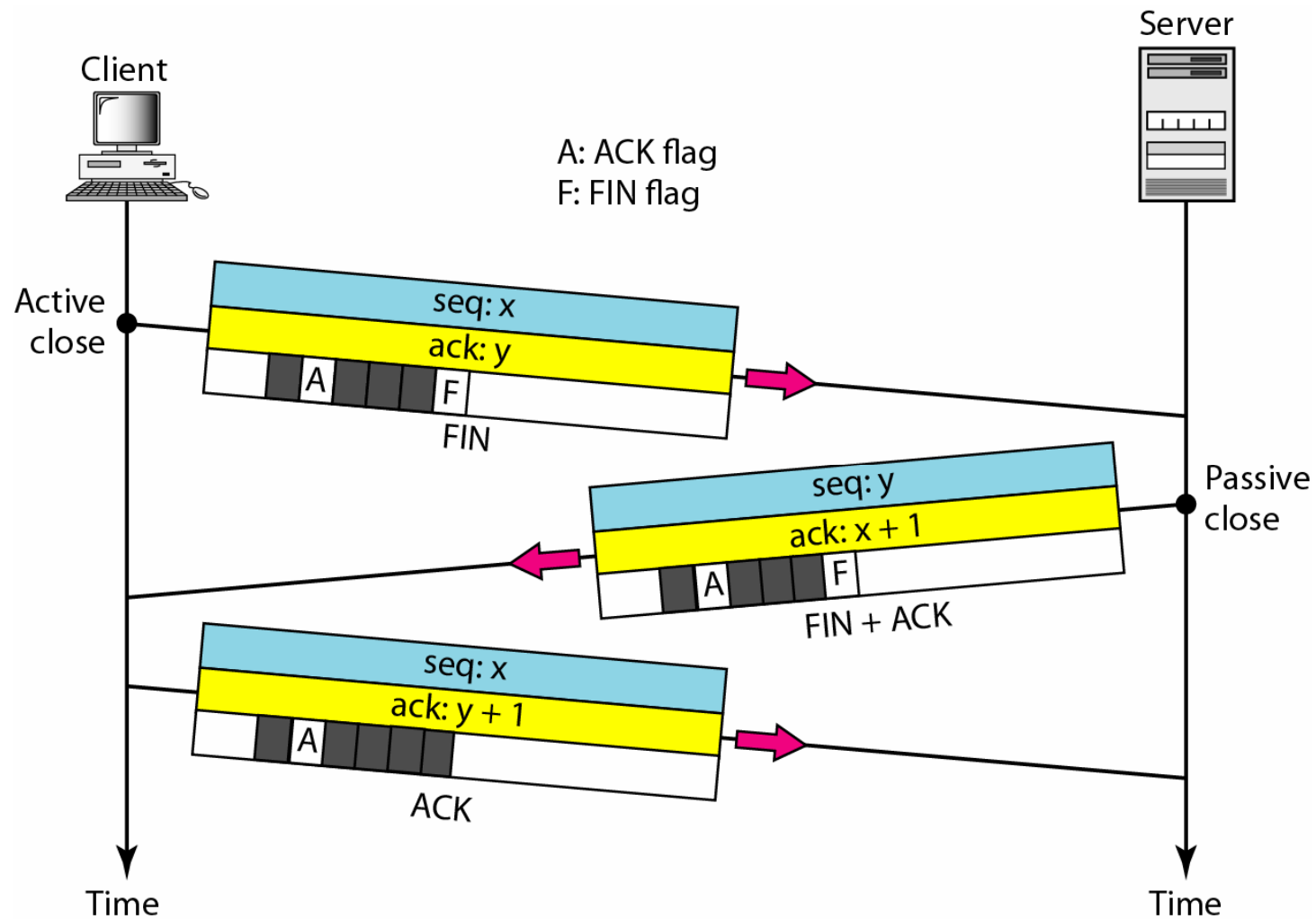


TCP data transfer



- The data transfer uses full duplex, piggybacking and Go-back-N.
- The receiver saves all segments that are received in the wrong order.

TCP connection termination



Domain Name System (DNS)

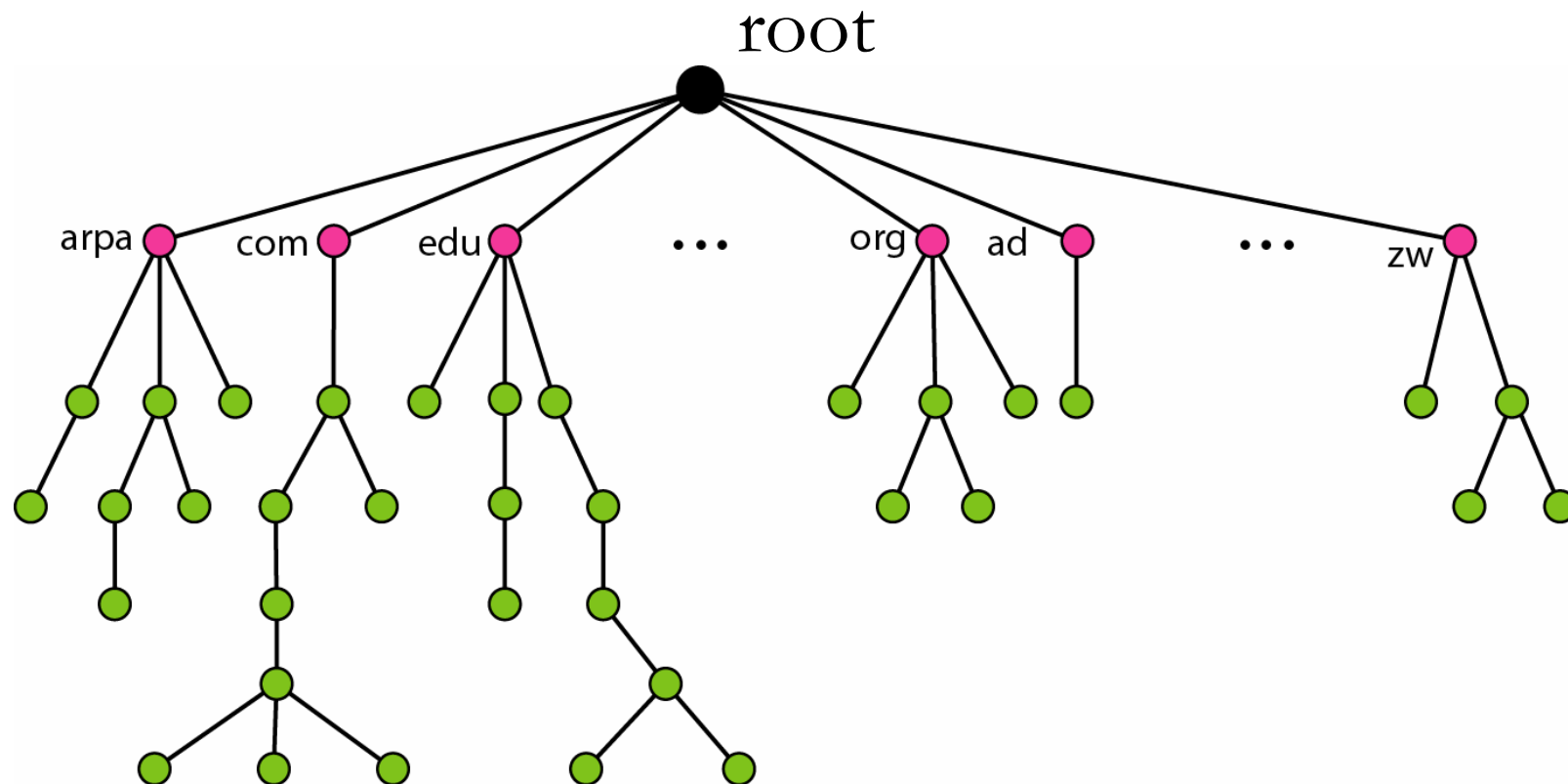
Many applications use symbolic names instead of numeric addresses, e.g Email addresses and URLs.

Domain Name System (DNS) is a supporting program that map a symbolic name to an address (and maps addresses to names).

DNS uses a **hierarchical name space** and the whole Internet is divided into **domains** and **subdomains**.

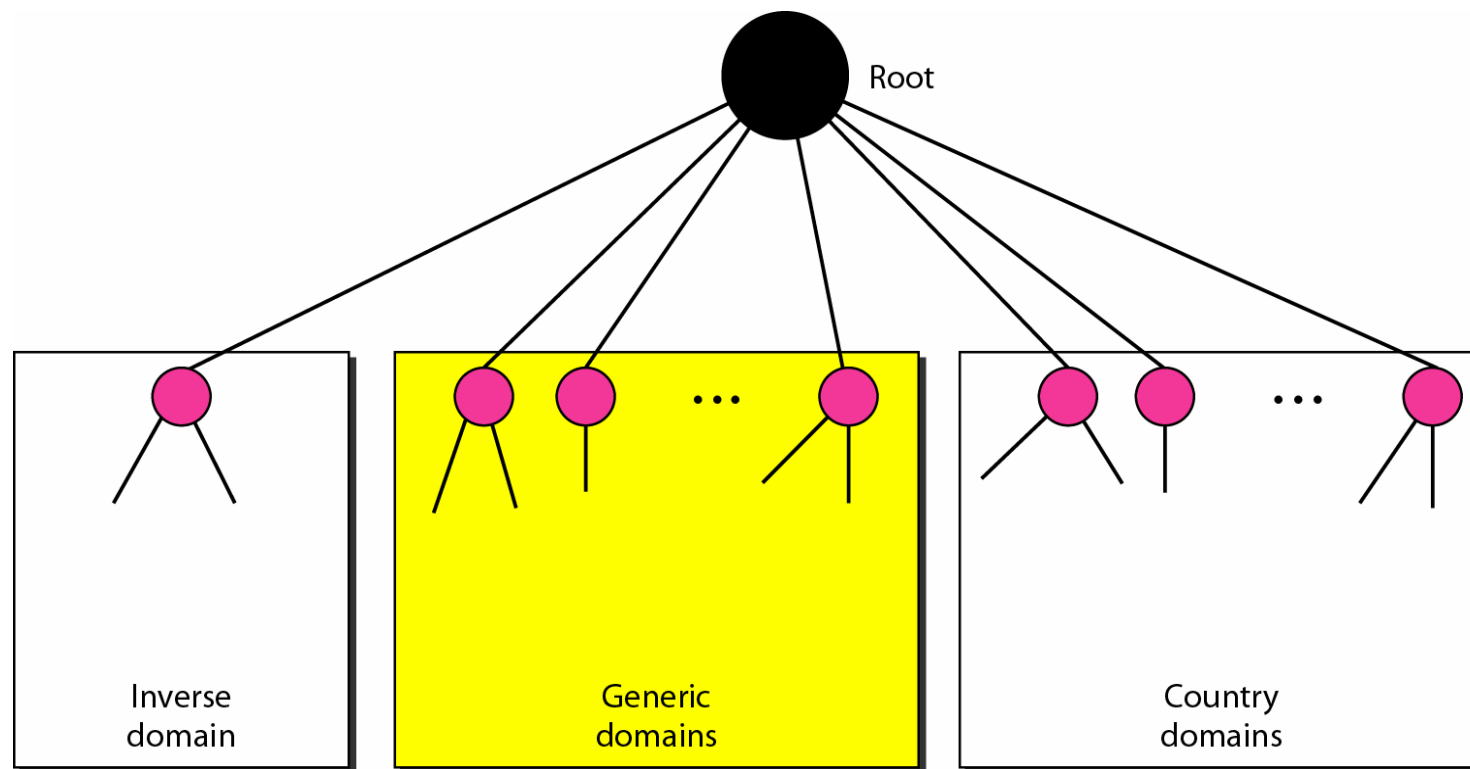
A **domain name** is a sequence of labels separated with dots, e.g. `www.telecom.lth.se`.

Domain name space in DNS

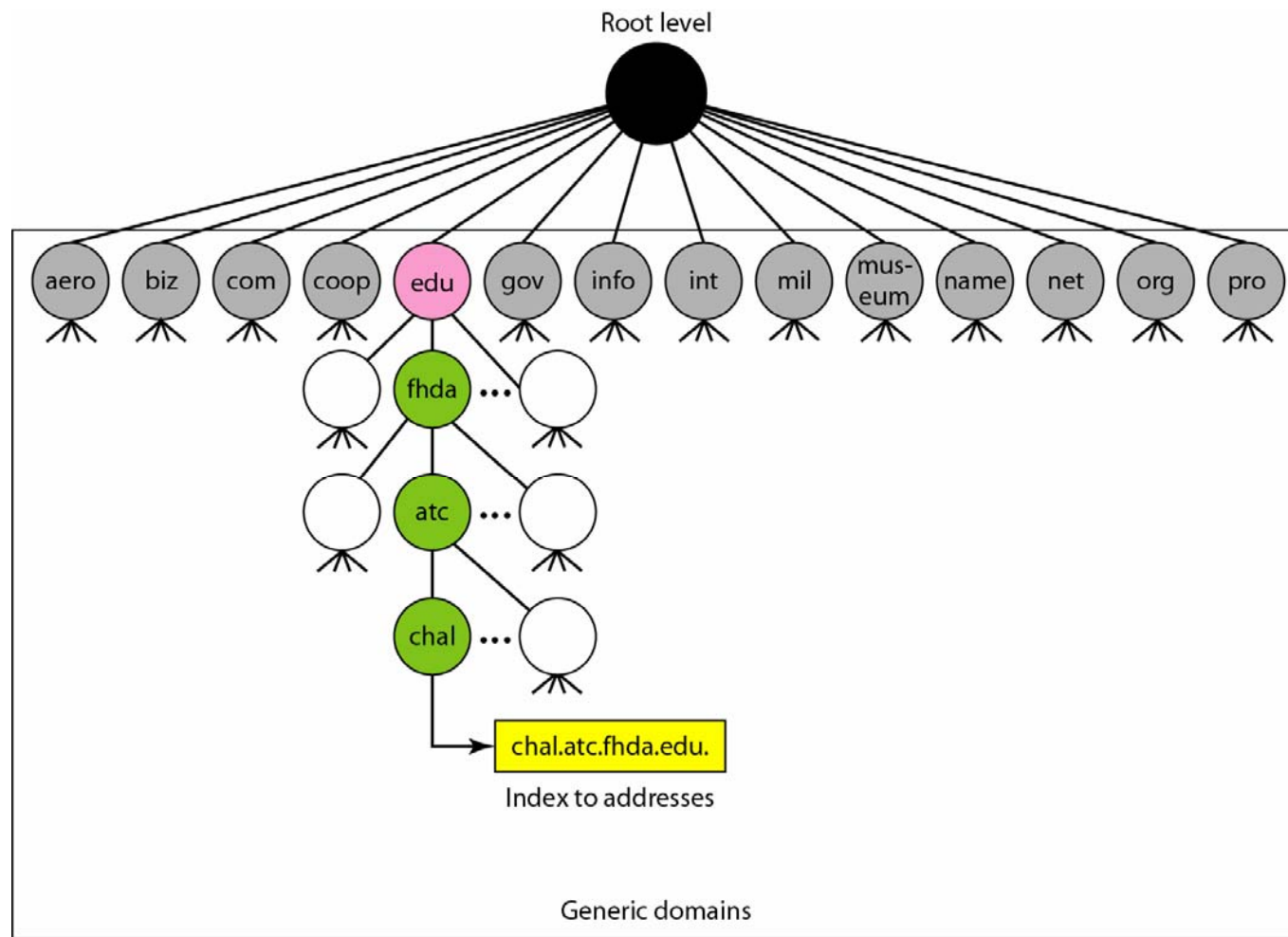


Domain types

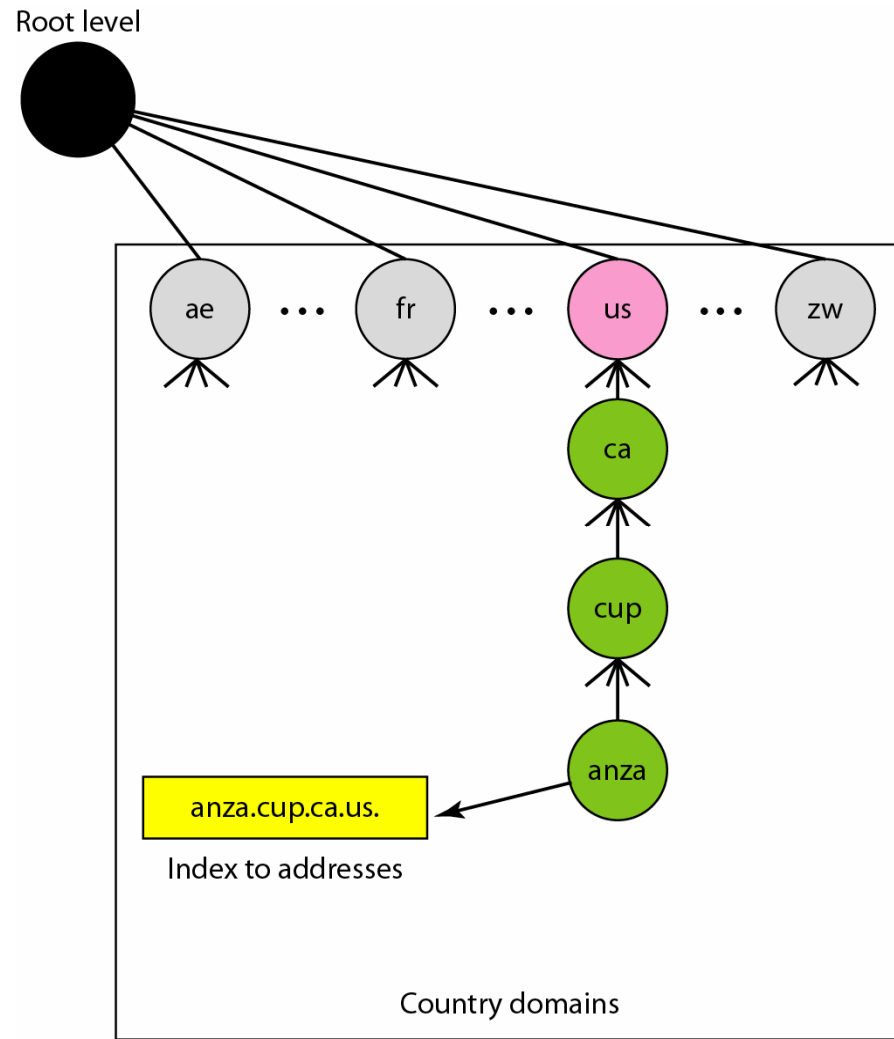
The domain name space is divided into three different sections.



Generic domains

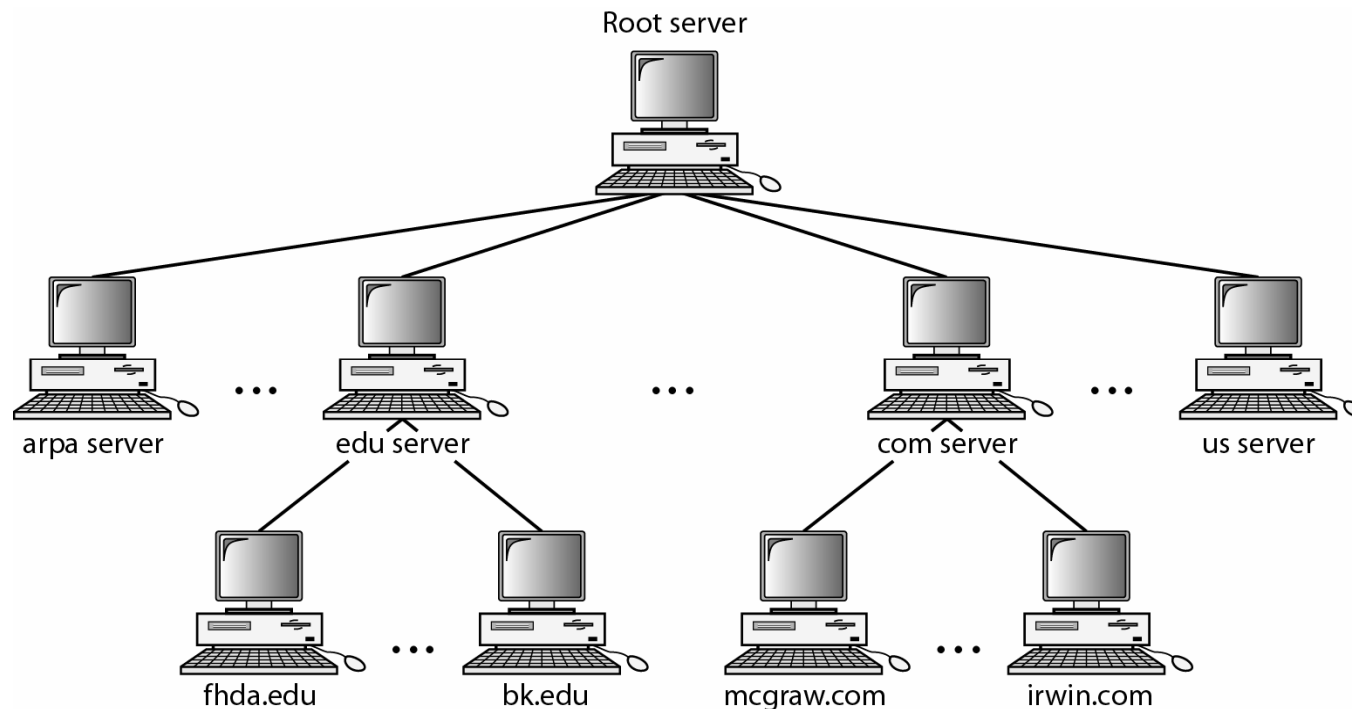


Country domains

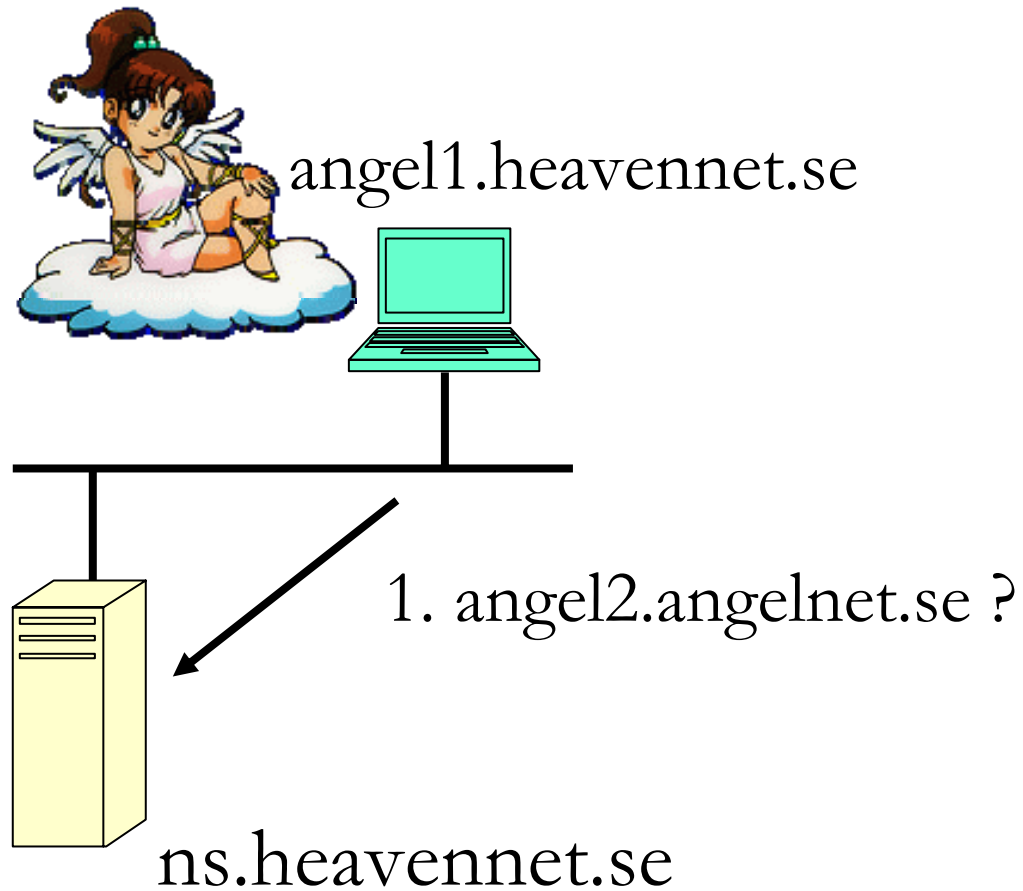


Domain Name Servers

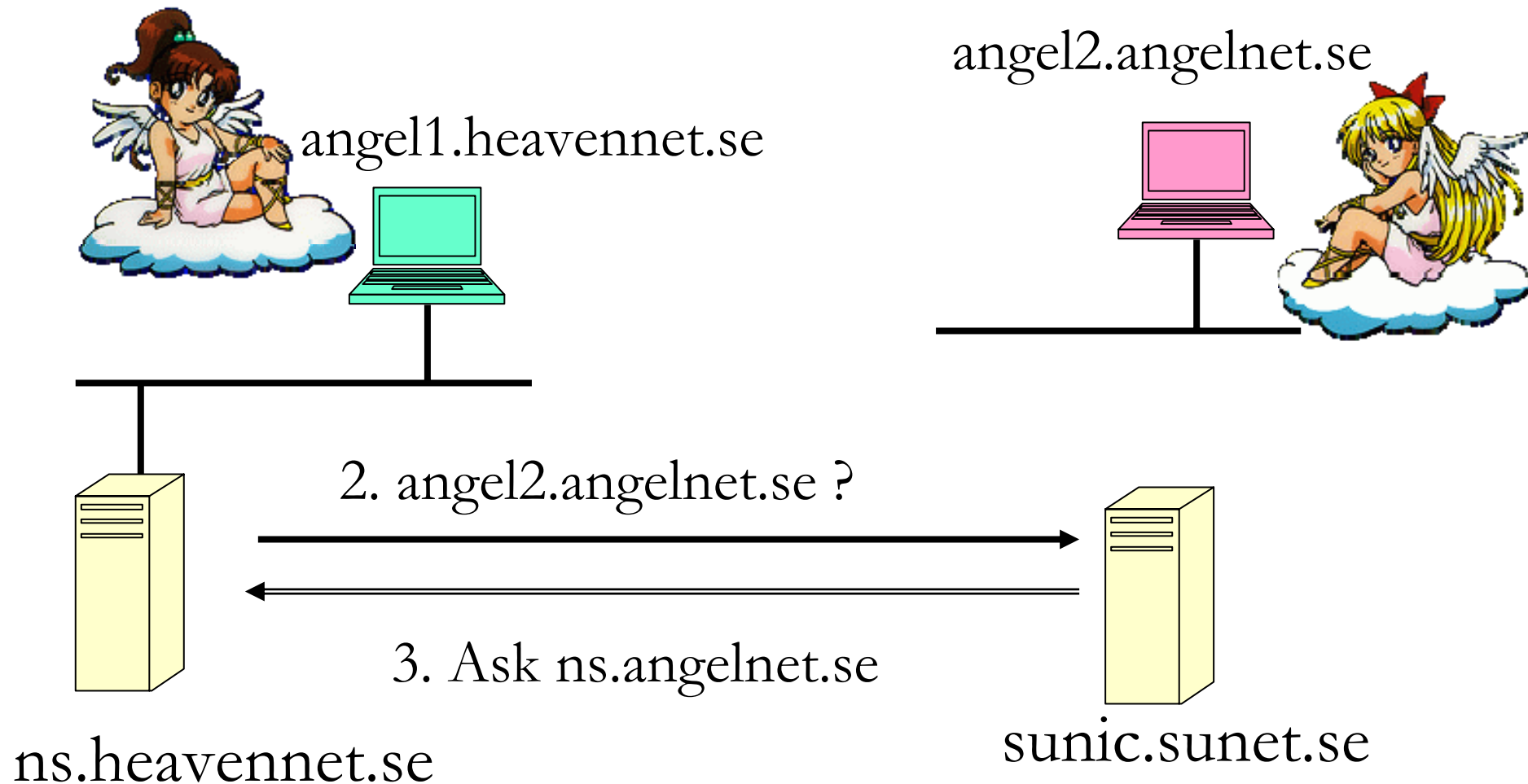
The domain name space is stored in special **DNS servers**. The servers are distributed and each domain or subdomain have their own servers.



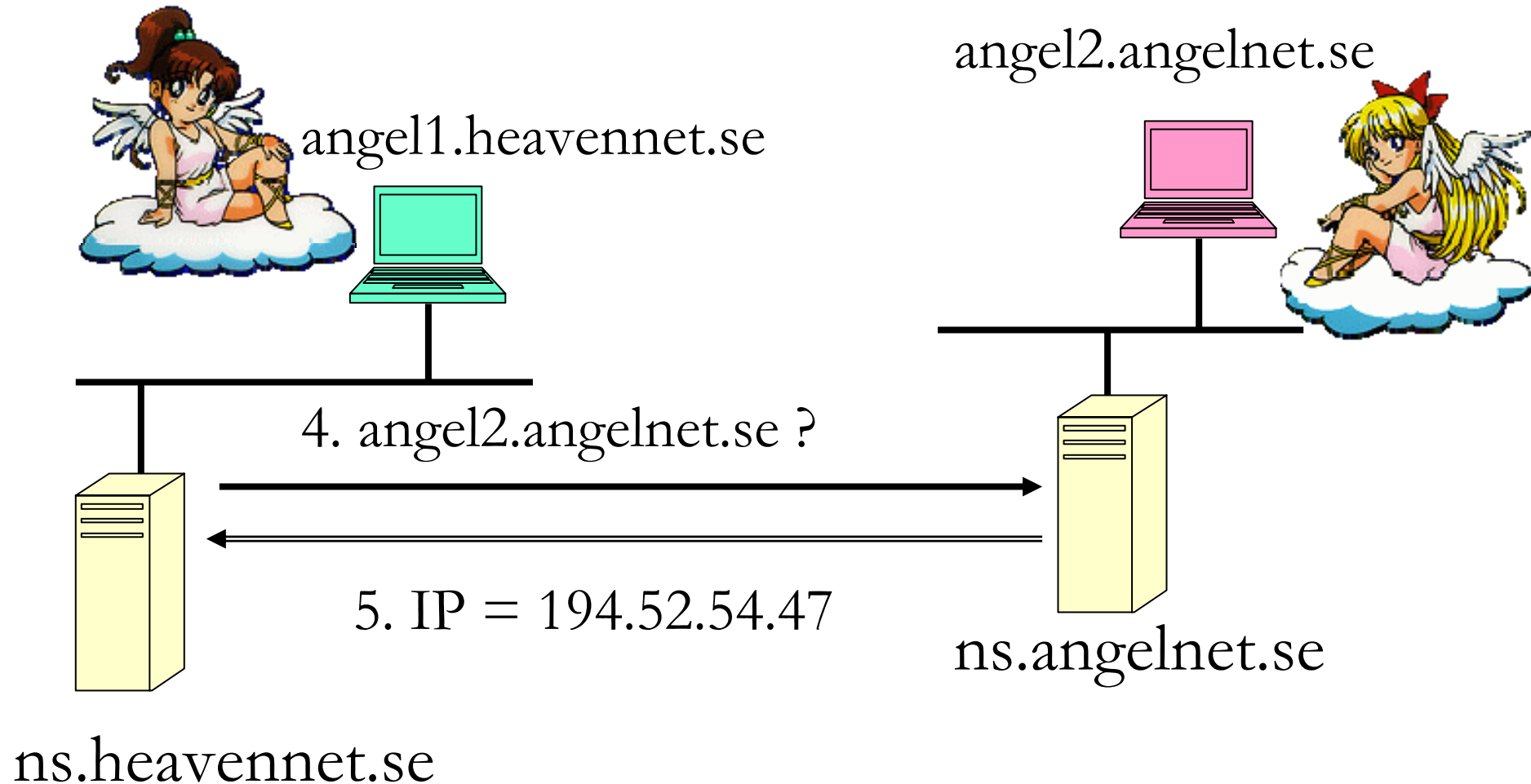
From name to address (1)



From name to address (2)



From name to address (3)



Address Resolution Protocol (ARP)

In order to find a specific host/router in a network, the IP-address must be mapped to a physical address.

Address Resolution Protocol (ARP) is used for the mapping of IP-addresses to MAC-addresses within a LAN.

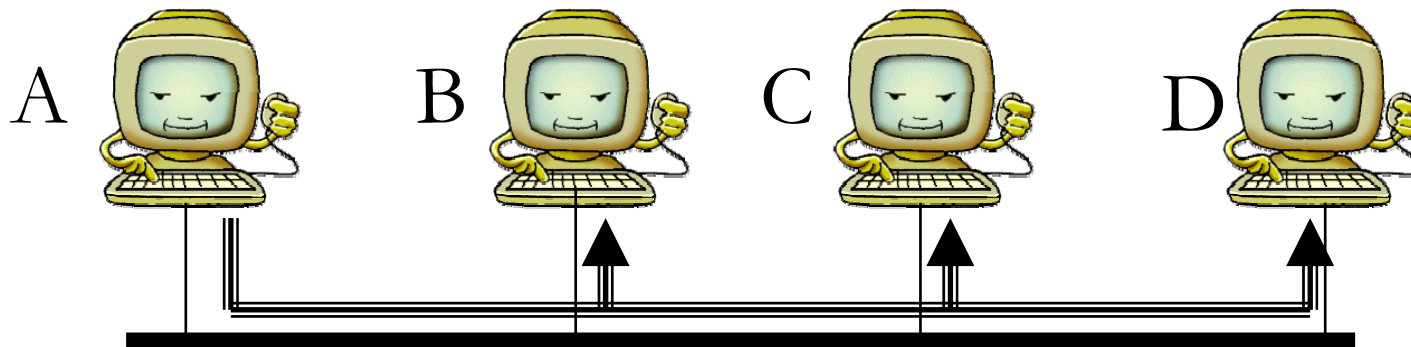
A host in a LAN always knows the IP-address to a default router (gateway) that is connected to the "rest of Internet".

ARP operation

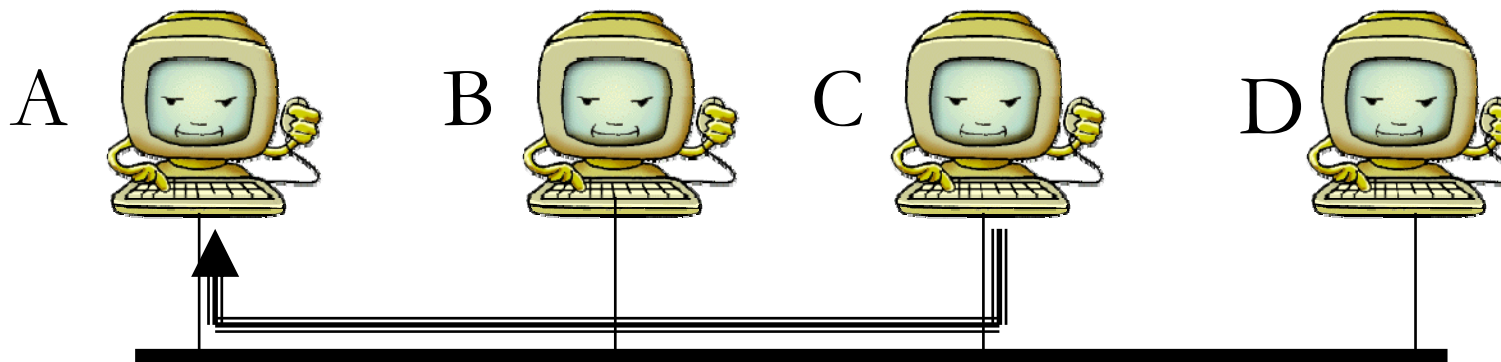
- Each host/router has a ARP cache (table) that is used to store MAC/IP-address pairs.
- An **ARP query packet** is broadcasted everytime a host/router needs to map an IP-address to a MAC-address.
- The intended host answers with an **ARP response packet**.

ARP, example

ARP query IP v.x.y.z



ARP response MAC a:b:c:d:e:f



ARP, example

If the intended host is on another network, the packet is sent to the default gateway.

